Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Plant-Based Antiviral Therapeutics for Livestock and Human Viral Infections			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Natural Pharmaceuticals	Industrial Biotechnology	Bio-based Agricultural/ Livestock/Aquatic preparations
Concept of Proposed Technology	**1. Project Background and Objectives** This project addresses the rising threat of zoonotic and emerging infectious diseases by developing plant-derived bioactive therapeutics, such as polyphenol-mineral complexes. The objective is to develop and formulate nanostructured, sustainable, low-toxicity, broad-spectrum antiviral agents. These solutions will provide scalable alternatives to synthetic antivirals by integrating botanical pharmacology with advanced nanobiotechnology and biomedical engineering. **2. Core Functional Mechanisms** The therapeutic agents utilize a multi-modal mechanism to inhibit viral progression:			

	• **Entry Inhibition:** Binding viral glycoproteins or host receptors to prevent cellular attachment and fusion. • **Replication Blockade:** Disrupting viral polymerases, proteases, or nucleic acid synthesis to halt replication. • **Assembly and Release Disruption:** Inhibiting virion maturation and egress while modulating cytokine responses to reduce systemic inflammation. **3. Applications and Clinical Scope** The technology is designed for dual-use across human and veterinary medicine: • **Veterinary Applications:** Feed additives and disinfectants targeting viral, bacterial, and fungal pathogens in livestock. • **Human Therapeutics:** Oral or topical formulations for pandemic preparedness against emerging RNA viruses (e.g., influenza, coronaviruses). • **Combination Platforms:** Adaptable formulations for use as standalone treatments or as adjuncts to conventional antiviral therapies.
Rationale for Joint R&D Project	The complexity of emerging worldwide viral threats (e.g., avian influenza virus) necessitates an integrated R&D approach that bridges laboratory discovery and field-scale application. This international collaborative initiative combines advanced bionanotechnological and biomedical expertise with industrial needs and deployment capacity to establish a comprehensive pandemic preparedness framework.
Key Words	1. Antiviral therapeutics 2. Polyphenol-mineral complexes 3. Spray nanocoating and administration 4. Feed additives 5. Plant-derived resources

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	This project aims to develop plant-derived antiviral therapeutics targeting key RNA/DNA viruses affecting livestock (e.g., PRRS, ASF, FMD, avian influenza) and humans (e.g., influenza, coronaviruses).			
R&D Target Specifications	**1. Targeted Technologies** • **High-Purity Materials:** Scalable production of plant-derived materials (e.g., tannins, flavonoids, and lectins), targeting ≥**95% purity** for feed and pharmaceutical applications. • **Advanced Formulation:** Nano-formulated systems (e.g., nanoemulsions, nanoparticles, liposomes) designed to stabilize plant-derived complexes and maximize efficacy. • **Field-Scale Validation Standards:** Real-time quantification of viral load reduction in livestock under field conditions. **2. Measurable Performance Standards** • **Antiviral Potency:** Achieve an IC_{50} ≤5 μg/mL • **Livestock Impact:** Achieve a ≥40% reduction in mortality for ASF/PRRS and maintain			

	≥90% additive stability/efficacy under thermal stress (40 °C for two months). • **Human Benchmarks:** Secure ≥80% inhibition of viral entry (e.g., coronavirus) with an EC_{50} ≤2 μM in human cell lines. The project will advance technologies from TRL 3 (Laboratory Validation) to TRL 6 (Pilot-Scale Demonstration), bridge the gap between bench research and industrial application.
R&D Plan	**36-Month R&D Roadmap** **Task 1: Material Optimization and Lead Identification (Months 1-18)** • **Focus:** Screen plant-derived materials and their complexes for antimicrobial capacity. • **Milestones:** Secure ≥5 cadidates; achieve IC_{50} ≤5 μg/mL; finalize lead compound selection. **Task 2: Mechanistic Analysis and Formulation (Months 12-36)** • **Focus:** Characterize antiviral pathways (entry/replication/assembly) and develop nano-formulated systems. • **Milestones:** Validate > 90% plaque reduction and ≥3 formulations. **Task 3: Preclinical Validation and Field Efficacy (Months 1-24)** • **Focus:** Conduct animal challenge studies and human cell assays; monitor viral load, survival, and production metrics. • **Milestones:** Achieve ≥70% viral reduction and ≥60% survival; demonstrate ≥40% mortality drop and ≥90% field stability. **Task 4: Scale-Up and Regulatory Preparation (Months 25-36)** • **Focus:** Execute pilot-scale production (10 kg batches), conduct toxicological profiling, and finalize IP/regulatory filings. • **Milestones:** Successful TRL 6 demonstration and completion of the final commercialization roadmap.
Expected R&R of Research Institute	The two academic research institutions, one in the USA and the other in Korea, will lead the scientific R&D in a synergistic and complementary mannar, providing the biomedical and nanobiotechnological expertise required for formulation development and validations of plant-derived antiviral complexes. Collectively (though not necessarily individually), they will contribute the following technical capabilities, resources, and infrastructure. **Technical Contributions** • **Formulation Development:** Identify nature-derived antiviral materials and complexes, and optimize formulations to maximize efficacy; develop nanocoating strategies for viruses and/or cells. • **Preclinical Validation:** Perform cytotoxicity profiling and neutralization assays in human cell lines. **Resources and Infrastructure** • **Facilities:** Provide access to BSL-2 (or 3) laboratories, analytical suites (LC-MS/HPLC, confocal microscopy, bioAFM), high-throughput screening platforms, and cell-culture facilities • **Personnel:** Assign dedicated principal investigators and specialized technical staff for the full 36-month project duration.

4. R&R (Roles and Responsibilities)

Expected R&R of Korean SME Partner	**Roles and Responsibilities: Industrial and Technical Partners** The project is supported by a consortium of specialized partners, enabling a seamless transition from laboratory research to large-scale field implementation. Academic formulation and validations teams in two countries will work synergistically to achieve the project objectives. **1. Lead Industry Partner: Application and Product Development** • **Field Operations:** Conduct large-scale on-farm challenge studies to validate efficacy in livestock populations (targeting ≥70% viral reduction). • **Industrial Scale-Up:** Manufacture high-purity (≥92%) food-grade materials in 10–100 kg batches. • **Product Formulation:** Develop stable and palatable feed additives suitable for routine livestock use, with a shelf-life of > 28 days. **2. Formulation Partner: Candidate Identification and Formulation Development** • **Candidate Identification:** Identify plant-derived antimicrobial candidates and supporting additives (≥5). • **Formulation:** Design nano-formulation systems for spray- and solution-based administration. • **Preclinical Testing:** Conduct cell-based assays to confirm microbial activity. • **Resources:** Leverage nanotechnology laboratories specialized in cell-based nanobiohybrid systems, together with advanced (bio)analytical instrumentation, to develop scalable prototypes. **3. Validation Partner: Formulation Development and Efficacy Evaluation** • **Formulation:** Develop nano-formulation systems with long-term stability/efficacy (≥90% at 40 °C for 2 months). • **Technical Testing:** Conduct high-throughput screening for potency and perform biochemical assays to confirm viral inhibition. • **Safety Profiling:** Perform cell-based toxicology assessments and biomarker analysis to ensure safety. • **Resources:** Provide BSL-2 facilities (preferably, BSL-3) to support TRL 4-6 milestones.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project will deliver validated plant-based antiviral therapeutics targeting livestock and human viruses, achieving TRL 6 prototypes with proven efficacy. **Technological Achievements** • Broad-spectrum antivirals: Scalable, food-grade formulations (e.g., tannic acid/EGCG nano-emulsions) with IC_{50} ≤5 μg/mL and ≥40% livestock mortality drop. • Integrated platforms: GLP-validated efficacy data and field-tested feed additives stable for two months at 40 °C. **Commercialization Potential** • Market-ready products: Veterinary feed supplements for PRRS/ASF (KRW 50-100B domestic market); over-the-counter human oral/topical drugs for influenza/coronaviruses via licensing. • IP & Scale-up: ≥2 patent applications on formulation and others; pilot manufacturing (100 kg/month) enabling 2-year commercialization post-TRL 6. **Broader Impacts** • Economic Benefits: Boost swine production, reducing Korea's estimated KRW 1-2T annual losses from ASF/PRRS.; • Industrial Ecosystem: Establish domestic supply chain for plant antivirals, enhancing food security and pandemic preparedness.

6. References

(1) G. Ameni, et al. A Narrative Review on the Pandemic Zoonotic RNA Virus Infections Occurred During the Last 25 Years. *J. Epidemiol. Glob. Health* **2024**, *14*, 1397.

(2) B. Bellido-Martín, et al. Evolution, Spread and Impact of Highly Pathogenic H5 Avian Influenza A Viruses. *Nat. Rev. Microbiol.* **2026**, *24*, 45.

(3) J. L. Kiley, et al. Avian Influenza: An Overview and Clinical Status. *Curr. Infect. Dis. Rep.* **2025**, *27*, 1.